

M. J. Biopharm Private Limited		
Plot No. 18, International Biotech Park, Hinjawadi, Phase II, Pune 411057.		
Format for Attachment		

Title: Outcome of Recommendation action -III of deviation occurred during calibration of TOC Analyzer (Instrument ID: I/QC/018)	
Attachment No: 	Ref. document No: DEV/QC/25/010

Outcome of Recommended Action III:

As per recommended action, repeat calibration activity planned on same day i.e. 21/03/2026 and work allotted to analyst Emp. Id. 952.

As per plan analyst Emp. Id. 952 verified all the pre requisites before initiation of preparation for 2-point calibration and System suitability test as per SOP, 'Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With Lab Solution Software, SOP No.: SOP/DP/QC/045.

After ensuring the availability of all prerequisites, Analyst initiated the preparation of Sucrose and 1,4 – Benzoquinone standard solution, under the supervision of QC reviewer Emp.ID 616 and details are as below;

Sucrose stock solution (50000 ppb):

Weighed accurately 11.892 mg of sucrose and transferred into the 100 mL volumetric flask which containing purified water, shake well and sonicated for 30 seconds to dissolve completely and diluted to 100 mL with purified water. Refer Attachment XVII & XVIII.

Sucrose standard solution (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with Purified water.

After completion of preparation, analyst filled the freshly prepared Sucrose standard solutions (500 ppb) and water in two individual 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray.

Further, initiated the run for 2-point calibration to generate the calibration curve and details of the generated curve as below; Refer Attachment XIX.

Name of Curve: CAL_21032026_01.2026_03_21_20_39_03.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	4.431	0.08021	1.81 %
Two Point Calibration (500 ppb)	36.82	0.8698	2.36 %

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Further calculated the TOC of blank water which is injected to establish the calibration curve and observed that obtained TOC of blank water is 68 ppb, which was within the acceptance criteria (NMT 100 ppb).

Since, obtained results for two-point calibration found well within the limit as per SOP/DP/QC/045, therefore analyst prepared the 1,4 -Benzoquinone standard solution and proceeded for 'System Suitability Test' (SST), the details of preparation are as below;

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weighed 7.510 mg of 1,4 -Benzoquinone & transferred in 100 ml volumetric flask, then added 50 ml of purified water in above 100 ml volumetric flask, sonicated for 30 seconds to dissolve completely, then make up the final volume to 100 ml. Refer Attachment XVII & XVIII.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with ultra-pure water.

Further, filled the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) in 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray and entire activity was performed by following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026_01.2026_03_21_20_39_03.cald'.

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

After completion of above run, verified the obtained results and noted that SST results obtained from freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) i.e. 98.73 % are found well within the acceptance criteria, the details are as below (Refer Attachment XX).

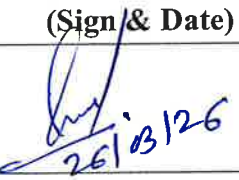
Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Freshly prepared 1,4 - Benzoquinone	98.73 %	36.41	0.5314	1.46 %
(SST Limit: Between 85-115 %)				

Based on the obtained results, it is observed that mean area sucrose (36.82) obtained from 2-point calibration and mean area of 1,4 – Benzoquinone (36.41) obtained from system suitability run found comparable with each other which led to % recovery found well within the acceptance criteria.

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Inference:

Based on the outcome of investigation and obtained results, it was confirmed that, during the initial calibration, sucrose and 1,4 – Benzoquinone standard solutions were prepared improperly or solutions were not mixed uniformly which led to variation in the area of sucrose and 1,4 – Benzoquinone consequently % recovery of the 1,4 – Benzoquinone found at higher side, however during the repeat calibration analyst prepared the standard solution properly by sonication and as an outcome all the obtained results found well within the acceptance criteria.

Prepared By (User) (Sign & Date)	Reviewed By (User) (Sign & Date)	Reviewed By (QA) (Sign & Date)
 26/03/26	 26/03/26	 26/03/26

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Root Cause Categorization

Deviation Number:

Date of Categorization:

Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
☒ Human Performance	The departure is due to an error made by a human	☒ Learning Error	The person lacks the knowledge and/or skills required to perform the task	☒ Not of curriculum
				☒ Thought they had been trained or didn't know how to verify training.
				☒ Not trained on latest version
				☒ Not trained on what could go wrong. Information not currently in a procedure or training materials.
				☒ Wrong method of training used for task/skill
				☒ Trained on irregular shift for learner (too tired)
				☒ Not capable of learning/understanding
				☒ Long delay between learning and use, or last time performing task
				☒ Lack of support and coaching for new learning
				☒ Responsibility boundaries non-existent /ineffective
		☒ Omission Error	The person has the knowledge and/or skills but misses a step/task/information or confuses items	☒ Process to remember inadequate
				☒ Know but missed - Physiological factors
				☒ Know but confused - Physiological factors/stressors controls non-optimal
				☒ Auto-Sequence unavailable or ineffective

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Sign/Date: 25/03/2026

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Root Cause Categorization

Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
		☒ Application Error	The person makes a mistake at the point of applying the knowledge and/or skills	☒ Auto-Sequence unavailable or ineffective ☒ Complexity ☒ Interruptions / switching ☒ Cognitive Processing for individual ☒ Transcription not minimized / non-optimal aids ☒ Physiological factors ☒ Ergonomic Issues non-optimal ☒ Equipment layout or design hard to operation
		☒ Inconsistency Error	The person believes they have applied the knowledge and/or skills correctly, but there is a lack of clarity or subjectivity in the Standard or Document.	☒ Visual Standard Lacking ☒ Physical Standard Lacking ☒ Document Standard Lacking ☒ Incorrect Standard Content ☒ Memory Distortion ☒ Physiological and Environmental Factors
		☒ Decision Error	The person makes a conscious choice with good intent, which results in an undesired outcome.	☒ Desired results are, or are perceived to be in conflict with other priorities. ☒ Desired results not communicated accurately and /or understanding was not verified ☒ Information used was not representative / adequate, relevant, accurate or current Consequences of decision was not known/understood

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Root Cause Categorization

Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
			or physical space controls for protecting systems or components were breached	
		☒ Maintenance inadequate ineffective	Due to issues with the equipment/instrument/facility/utility, the current maintenance schedule or protocol is inadequate. This is not caused by human error- but due to inappropriate maintenance schedule.	NA
		☒ Mechanical Failure	Issues with the equipment/instrument/facility/utility leading to mechanical failure. This is not caused by human error- but by issues with the items being maintained.	NA
		☒ Electrical Failure	The departure is caused by failure of the equipment/facility/utility due to electrical issues.	NA

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☒ Material	The departure is due to a problem inherent to the materials and/or components	☒ Lack of Process Knowledge	Process is new or new to facility and knowledge of process is incomplete at this point in time	NA
		☒ Defective material	The material is not suitable for use.	NA
		☒ Incorrect material	The wrong material is present.	NA
		☒ Damaged material	The material is correct, was not defective when received, but was damaged by physical or chemical means so that is no longer appropriate for its intended purpose.	NA
		☒ Supplier Quality Issue	Supplier Quality Issue	NA
☒ Environment	Departure is due to an alert/action level excursion in regards to environmental and utility monitoring or due to outside	☒ Outside Environment	Weather conditions are outside the normal, expected parameters and caused systems to fail to operate as per intended purpose. The Equipment /Instrument/facility/utility is set	NA

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25 MAR 2026

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Root Cause Categorization

Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
☒ Equipment/Instrument		☒ Lack of Process knowledge	Process is new or new to facility and knowledge of process is incomplete at this point in time.	NA
☒ Equipment/Instrument	The root cause is due to an issue with Equipment/Instrument/Utility that is not caused by a human error.	☒ Facility/Equipment/Instrument/Utility Design	Design of the Facility/Equipment/Instrument/Utility is inadequate or inappropriate; for example the specification for a product is tighter than the tolerance that an equipment making the product is capable of meeting.	NA
	Equipment/Instrument/Utility does not perform as expected	☒ Computer System	Configuration errors, master data errors or software code errors caused by the incident. Validation testing or change control test plans did not test for problem identified. Partial or total failure of the computer system to operate. Software security issues related to use of electronic signatures, user ID or passwords. Technical	NA

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Root Cause Categorization

Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
☒ Method / Procedure	The departure is due to an issue with an unclear/lacking SOP/Test Method/Batch Record or other GMP document that gives direction and/or guidance on performing tasks.	☒ Document detail	The document that has the directions to do the task correctly or information that provides required knowledge is missing information.	☐ Impact of deviations /variance in process / task not known. Emotional and/or experiential (Limbic/gut feeling) response led to wrong decision ☐ Strength, probability and/or timing of consequences are not appropriate to drive correct decision. ☐ Behavioral standards are being ignored to preserve "relationships" ☐ Inappropriate behaviors/decisions are wide-spread within the team.
		☒ Document	The document has incorrect information.	NA
		☒ Validation/qualification inadequate	The Validation/Qualification that was done was not adequate.	NA
		☒ Process variability determined	The process has inherent variability which is causing the problem.	NA

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Primary Root Cause Categories	Definition	Secondary Root Cause (Level 2)	Definition	Tertiary Root Cause: (Level 3)
	environment (e.g. weather) or work place configuration.		up correctly but cannot cope with the unexpected extremes.	
		☒ Viable/non-viable particles inherent process	Alert/Action level excursion (control limit) is due to an inherent process related activity.	NA
		☒ Internal work area layout inadequate	Internal work area layout inadequate	NA

Remark (If Any)	NA
Primary Root Cause	Method / procedure.
Secondary Root Cause	Document unclear / Lacking detail
Tertiary Root Cause	NA
Checked By (User)	 26/03/26
Reviewed By (QA)	 26/03/26

Reference SOP No.: SOP/DP/QA/005

Unit: Drug Product

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Training Documentation Form

Training Documentation Form

Topic of Training		Awareness training against deviation		Venue: <u>02</u>	
Document No.		<u>DEV/02/25/010</u>		Revision No.: <u>NA</u>	
Date		<u>25/03/26</u>		Time: <u>13:32 to 13:41</u>	
Type of Training (Tick ✓ the appropriate)		Tier I ☒ Tier II ☒ Tier III ☒ Tier IV ☒			
		Classroom ☐ Shop Floor ☒ On job training ☒ Refresher training ☒			
		Sign/date		Employee code	
		Name of the Trainee		Sign/date	

[illegible]

Name of Trainer:	Designation and department	Sign/date:
Pravin Kolhand	Asst. Manager and AC	 25/03/26

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